

1. Difference Between a Virus and a Worm

Aspect	Virus	Worm
Definition	A malicious program that attaches itself to a host file or program and requires user action to spread.	A standalone malicious program that self-replicates and spreads independently across networks.
Spreading Mechanism	Needs a host file or program; spreads when the infected file is executed.	Spreads automatically without needing a host or user interaction.
Activation	Usually requires user action (e.g., opening a file).	Can spread automatically once executed.
Payload	Can damage files, delete data, or deliver other malware.	Primarily focused on rapid spreading; may carry payloads.
Impact on Network	Less focused on network; mainly local infection.	Can cause network congestion due to rapid propagation.

2. Types of Viruses

- **File Infector Virus:** Attaches to executable files and activates when the file is run.
- **Boot Sector Virus:** Infects the boot sector of a hard drive or removable storage, loading before the OS.
- **Macro Virus:** Targets macro languages in applications like MS Word or Excel.
- **Polymorphic Virus:** Changes its code or appearance with each infection to evade detection.
- **Resident Virus:** Embeds itself in memory and intercepts system operations to infect files on the fly.

- **Multipartite Virus:** Infects multiple parts of a system (files and boot sectors).
 - **Stealth Virus:** Hides its presence by intercepting system requests and presenting clean versions of infected files.
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3. Antivirus Evasion Techniques

- **Polymorphism:** Virus changes its code pattern to evade signature detection.
 - **Encryption:** Virus encrypts its body and decrypts only during execution.
 - **Code Obfuscation:** Making the virus code complex and hard to analyze.
 - **Rootkits:** Hide virus presence by manipulating OS functions.
 - **Packing:** Compressing/encrypting the virus payload inside a wrapper.
 - **Anti-debugging and Anti-VM:** Detect if running in a sandbox or debugger to avoid analysis.
 - **Metamorphism:** Virus rewrites its own code completely each time it infects.
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4. Virus Detection Methods

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Signature-based Detection: Matches virus against a database of known virus signatures.

- **Heuristic Analysis:** Detects unknown viruses by analyzing code behavior and structure.
- **Behavioral Detection:** Monitors programs for suspicious behavior during execution.
- **Sandboxing:** Runs programs in isolated environments to observe malicious behavior.
- **Integrity Checking:** Compares file hashes or checksums to detect unauthorized changes.
- **Cloud-based Detection:** Uses cloud servers for real-time virus detection updates.